Hesam Salmabadi

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Google Scholar | ResearchGate | GitHub | LinkedIn

Experience

Geospatial Data Scientist May 2025 – Present
Agriculture and Agri-Food Canada (AAFC), Earth Observations, Science & Technology Branch Ottawa, ON, Canada (Remote)

- Assembled a national LULC data cube at 100 m resolution from multi-source satellite time series and environmental covariates covering all of Canada south of 60°N; engineered out-of-core ingestion and harmonisation pipelines (xarray, dask, Zarr) to support continental-scale spatial analysis
- Validated satellite-derived agricultural land estimates against Statistics Canada Census data across all 10 southern provinces; identified three distinct regimes of agricultural change — active conversion to urban land, gradual abandonment on marginal land, and administrative flux — with strong EO–Census agreement (Pearson $r = 0.988$)
- Developed Bayesian provincial demand forecasting models (PyMC) for LULC trajectories, reporting posterior medians and 94% HDI to quantify projection uncertainty; translated posteriors into five policy-relevant land-use demand scenarios
- Built pixel-level logistic regression suitability classifiers with geographic block cross-validation to prevent spatial leakage across disjoint test regions (ROC-AUC 0.957, Recall 0.81 on geographically held-out blocks)
- Ran Dyna-CLUE with demand scenarios and pixel-level suitability surfaces to generate spatially explicit LULC projection maps across Canada

PhD Researcher May 2021 – May 2026
Université du Québec à Trois-Rivières (UQTR) & Centre d'Études Nordiques, Université Laval Trois-Rivières, QC, Canada

- Recast the Soil Freezing Characteristic Curve (SFCC) into permittivity–temperature space, eliminating moisture-specific calibration; applied Bayesian hierarchical partial pooling across 87 in situ stations (10 networks, boreal, prairie, and tundra) — mean R^2 = 0.95; demonstrated that the zero-curtain transitional state persists $>$100 days in moisture-rich boreal sites, invisible to standard binary classifiers
- Developed a probabilistic non-binary freeze–thaw retrieval fusing SMAP (L-band) and AMSR2 passive microwave observations into a continuous Frozenness Index and 3-class state map at 9 km resolution across the Northern Hemisphere (10-year record); built end-to-end ingestion and continental-scale processing pipelines via Dask and Zarr; validated against a three-product unanimous consensus — match rates 78.8–98.2%, Cohen's Kappa 0.44–0.87, PR AUC $>$0.95 in three open biomes; applied DBSCAN clustering and StratifiedGroupKFold to eliminate spatial autocorrelation leakage
- Applied XGBoost with SHAP attribution over a 9-year continental record; identified sub-canopy snowpack as the primary driver of frozen ground duration in forested biomes — a regime shift invisible to temperature-only models (R^2 = 0.894, MAE = 11.5 days); linked the Frozenness Index to AmeriFlux ecosystem respiration via piecewise regression, confirming the zero-curtain onset marks the metabolic shutdown tipping point for cold-season carbon flux
- Conducted field campaigns across remote Canadian ecosystems; responsible for sensor installation, data logger servicing, and in-field quality control across boreal, subarctic, and prairie terrain
- Funded by FRQNT Doctoral Research Scholarship ($75,000 CAD, 2023–2026), UQTR Bourse Universalis Causa ($2,100 CAD), and Canadian Space Agency Research Grant ($\sim$$30,000 CAD via supervisor)

Geospatial Data Analyst — Contributing Author Apr 2024 – Oct 2024
Environment and Climate Change Canada (ECCC) Toronto, ON, Canada (Remote)

- Developed a multi-decade satellite data pipeline processing 40+ years of daily freeze–thaw observations into structured xarray spatial time series across Canada, underpinning the national-scale cryosphere trend assessment
- Quantified a reduction of 1.7 days/decade in frozen ground days since 1979 and a 28% increase in unfrozen days across the Canadian Tundra (unfrozen season lengthened by $\sim$27 days); produced publication-ready spatial trend maps for the federal climate assessment
- Contributing author, *Canada's Changing Climate Report 2026*, Chapter 6: Cryosphere Changes (forthcoming)

Field Research Technician Aug 2022
NASA Jet Propulsion Laboratory (JPL) — SMAPVEX22-Boreal Campaign Saskatchewan, Canada (On-site)

- Contributed to ground-based soil moisture and temperature measurements across the BERMS study area as part of a NASA SMAP satellite validation campaign; field data are part of Berg et al. (2025), *IEEE JSTARS*

Environmental Remote Sensing Scientist Oct 2019 – Jan 2021
Research Institute of Cultural Heritage & Tourism (RICHT) Tehran, Iran

- Designed an automated Lagrangian simulation pipeline (Python/PySPLIT) executing forward trajectories over 18 years to map altitude-dependent transport of salt and dust sediments from Lake Urmia's desiccating playa
- Engineered a multi-decadal satellite processing pipeline (MODIS AOD, Landsat) to track lake desiccation and identify active saline dust sources; established a strong correlation ($R = 0.86$) between exposed lake bed area and regional aerosol events
- Quantified a 76% loss in lake surface area linked to agricultural expansion; demonstrated that low-altitude dust transport drove a 30–60% increase in dust hours for downwind cities

MSc Researcher Sep 2015 – Jul 2018
Iran University of Science and Technology (IUST) Tehran, Iran

- Investigated the air quality impact of wetland desiccation in the Middle East; combined multi-decadal Landsat time-series for wetland area monitoring with HYSPLIT forward trajectory modelling — identified four primary dust transport pathways from the Mesopotamian Marshes, with regional implications for air quality from the Persian Gulf to Central Asia
- Monitored national SO_2 column concentrations over Iran (2005–2016) using Aura/OMI satellite retrievals; characterised seasonal transport corridors from an industrial point source via HYSPLIT forward trajectories
- Contributed geospatial analysis to a collaborative environmental study on heavy metal (Hg) distribution in Musa Estuary; produced spatial contamination maps using ArcGIS and geostatistical interpolation
- Produced 5 peer-reviewed publications (2019–2025); graduated as top-ranked MSc student (GPA 18.33/20, national scholarship)

Education

Doctor of Philosophy — Environmental Sciences May 2021 – May 2026
Université du Québec à Trois-Rivières (UQTR) Trois-Rivières, QC, Canada

Thesis: Non-Binary Monitoring of Seasonally Frozen Ground: From In Situ Sensors to Continental Passive Microwave Retrievals
Supervisor: Prof. Alexandre Roy

MSc — Civil Engineering (Environmental Engineering) Sep 2015 – Jul 2018
Iran University of Science and Technology (IUST) Tehran, Iran

Thesis: Determining the Effect of Wetland and Lake Desiccation in Iran on the Air Quality of Potentially Affected Regions
GPA: 18.33/20 — First Ranked Student *Supervisor:* Prof. Mohsen Saeedi

BSc — Civil Engineering Sep 2010 – Sep 2014
University of Birjand Birjand, Iran

Selected Publications

14 publications total (9 first-authored) | Citations: 181 | h-index: 6 | i10-index: 5 [Google Scholar]

Salmabadi H., Pardo Lara R., Berg A., Mavrovic A., Hanes C., Montpetit B., Roy A. (2026). In situ monitoring of seasonally frozen ground using soil freezing characteristic curve in permittivity–temperature space. *The Cryosphere*, 20, 1635–1654. doi:10.5194/tc-20-1635-2026

Salmabadi H., Berg A., Roy A. (2026, preprint). Non-Binary Retrievals of Soil Freeze–Thaw Dynamics Using Multi-Frequency Passive Microwave Data. *SSRN*. Under review. ssrn.com/abstract=6626104

Salmabadi H., Saeedi M., Notaro M., Roy A. (2025). Dust transport pathways from the Mesopotamian Marshes. *Aeolian Research*, 73, 100975. doi:10.1016/j.aeolia.2025.100975

Salmabadi H., Saeedi M., Roy A., Kaskaoutis D.G. (2023). Quantifying the contribution of Middle Eastern dust sources to PM_{10} levels in Ahvaz, Southwest Iran. *Atmospheric Research*, 295, 106993. doi:10.1016/j.atmosres.2023.106993

Salmabadi H., Roy A., Berg A. (2026). Capturing the Soil Zero-Curtain Effect from Multi-Frequency Passive Microwave Retrievals. *ISPRS Congress 2026.*

Berg A.A., ..., **Salmabadi H.**, ...(2025). Soil Moisture Active/Passive Validation Experiment Within the Canadian Boreal Forest in 2022 (SMAPVEX22-Boreal). *IEEE J. Sel. Topics Appl. Earth Obs. Remote Sens.*, 18, 11816–. doi:10.1109/JSTARS.2025.3564195

Skills

Programming & Data: Python (xarray, dask, numpy, pandas, rasterio, geopandas, scikit-learn, matplotlib, cartopy, h5py), GDAL (Python & CLI), Bash, MATLAB, LaTeX, Git/GitHub, MySQL

Machine Learning & Statistics: Logistic regression (L1/L2), random forests, SVM, DBSCAN, Bayesian hierarchical partial pooling, probabilistic classification, Mann-Kendall trend test, spatial cross-validation

Remote Sensing: SMAP (L-band), AMSR2, MODIS AOD, Landsat time-series, OMI SO_2, ERA5, GDAS; passive microwave freeze–thaw retrieval; source apportionment (PSCF)

Modelling: HYSPLIT trajectory modelling (forward & backward); numerical atmospheric transport; soil freeze–thaw modelling

Data Formats: NetCDF, HDF4/5, GeoTIFF, Zarr, Shapefile

GIS: QGIS, ArcGIS/ArcMap, PostGIS (learning), spatial interpolation (IDW, kriging concepts)

Cloud & Infra: AWS (in progress), GitHub Actions basics, agentic AI tooling

Awards & Funding

FRQNT Doctoral Research Scholarship 2023–2026
Fonds de recherche du Québec – Nature et technologies $75,000 CAD
Project: *Development of a New Passive Microwave Multi-Sensor Freeze/Thaw Data Product to Improve Growing Season Length Monitoring over North America*

Bourse Universalis Causa 2021–2024
Université du Québec à Trois-Rivières Up to $21,000 CAD + tuition exemption

First Ranked Student — MSc Program 2018
Iran University of Science and Technology (IUST) GPA: 18.33/20

Full Tuition Scholarship — BSc & MSc 2010–2018
Ministry of Science, Research and Technology of Iran — awarded on national competitive examination ranking

Certificates & Professional Development

- AI for Earth Program — Amii (Alberta Machine Intelligence Institute), 2026
- Advanced Learning Algorithms — DeepLearning.AI / Coursera, 2025
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI / Coursera, 2025
- Google Data Analytics Certificate (Foundations, Python, Data Exploration) — Google / Coursera, 2022–2025
- Advanced Python Programming — Maktabkhooneh, 2020

Languages

Persian (Native) | English (Fluent)